Computational Physics

Assessment 2 out of 3

April 21, 2026

Duration: 90 mins

Total Marks: 25

Useful Formulae:

1. Newton-Raphson method:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

2. Lagrange's interpolation:

$$P(x) = \sum_{i=0}^{n} y_i \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}.$$

3. Trapezoidal rule:

$$\int_a^b f(x)dx = \frac{h}{2}[f(x_0) + f(x_1)] = \frac{b-a}{2}[f(b) + f(a)].$$

4. Explicit Euler:

$$y_{n+1} = y_n + hf(t_n, y_n).$$

5. Runge-Kutta method order 4:

$$\begin{aligned}
y_{n+1} &= y_n + \frac{h}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right), \\
t_{n+1} &= t_n + h, \\
k_1 &= f(t_n, y_n), \\
k_2 &= f\left(t_n + \frac{h}{2}, y_n + k_1\frac{h}{2}\right), \\
k_3 &= f\left(t_n + \frac{h}{2}, y_n + k_2\frac{h}{2}\right), \\
k_4 &= f(t_n + h, y_n + hk_3).
\end{aligned}$$

6. Implicit Euler:

$$y_{n+1} = y_n + hf(t_{n+1}, y_{n+1})$$

Note: All questions are mandatory.

Questions:

1. (a) The positive zero of the polynomial $f(x) = x^2 - 4$ is determined using the Newton-Raphson method, using an initial guess $x = 1$. Let the estimate, after two iterations, be $x^{(2)}$. Calculate the percentage error $\left|\frac{x^{(2)}-2}{2}\right| \times 100\%$. [2]

   (b) Given the data points

x	1	3	5
y	4	28	92

   Using Lagrange's method of interpolation, determine the value of $y(4)$. [2]

2. (a) Write down the forward and backward finite difference method for the second-order derivative of a function $f(x)$, i.e.,$\partial^2 f(x)/\partial x^2$ with the second-order accuracy $O(h^2)$. [2]

   (b) Do the Taylor expansion for the forward finite difference method and show the order of accuracy for this case. [3]

3. (a) Derive the trapezoidal rule using the Lagrange linear interpolating polynomial. [3]

   (b) Do the error analysis step by step and provide the final form of error. One may follow the same procedure that we have used in our class. [3]

4. Consider the 1D advection–diffusion equation in conservative form:

$$\frac{\partial c}{\partial t} = \frac{\partial}{\partial x}\left(D\frac{\partial c}{\partial x} - vc\right)$$

   with parameters:

   - $D = 0.02, \quad v = 1$
   - Domain: $x \in [0, 1]$
   - Boundary conditions: $c(0, t) = c(1, t) = 0$
   - Initial condition:

$$c(x, 0) = 10 \exp\left(-\frac{(x - 0.5)^2}{0.02}\right)$$

   (a) What is the largest Δx required to resolve the initial profile? Justify your answer. [2]

   (b) Write the stability conditions due to advection (CFL condition) and diffusion. Which condition is more restrictive? For $\Delta x = 0.002$, determine the largest allowable time step Δt for stability using an explicit scheme. [3]

5. Consider a linear first-order ODE with a decay term:

$$\frac{dy}{dt} = -100y, \qquad y(0) = 1.$$

   Let ~~us~~ say h is the time step that one will use for the numerical time integration.

   (a) If we choose a very small time step $h = 10^{-6}$, and if we have an option to choose between explicit Euler and explicit RK4 with limited computational resources, which one will you choose and why? Justify your choice. [2]

   (b) If we choose $h = 2$, which one will perform better, implicit or explicit Euler? Will anyone of the two be stable? [2]

   (c) For $h = 2$, will explicit RK4 be stable? [1]